

Speak Like A Mathematician:

parallel lines:

perpendicular lines:

skew lines:

parallel planes:

transversal:

Parallel and Perpendicular Postulates

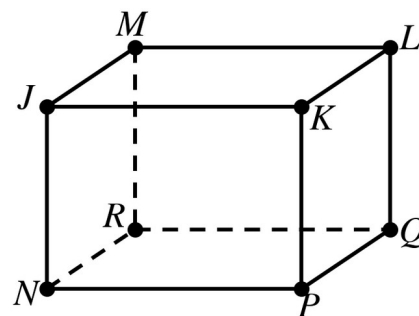
Parallel Postulate: If there is a line and a point not on the line, then there is exactly _____ line through the point _____ to the given line.

Perpendicular Postulate: If there is a line and a point not on the line, then there is exactly _____ line through the point _____ to the given line.

Example 1 - Lines, Segments, and Planes

Use the rectangular prism. Think of each edge as part of a line.

- Name a segment parallel to $\overline{JK}$.
- Name a segment skew to $\overline{JK}$.
- Name a segment perpendicular to $\overline{JK}$.
- Name a plane parallel to plane JKPN.



Angle Pairs Formed by a Transversal

Two angles are _____ angles if they lie in matching corners at the two intersections.

Two angles are _____ angles if they lie between the two lines on opposite sides of the transversal.

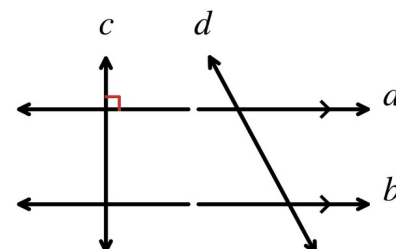
Two angles are _____ angles if they lie outside the two lines on opposite sides of the transversal.

Two angles are _____ angles if they lie between the two lines on the same side of the transversal.

Example 2 - Parallel, Perpendicular, or Neither

Use the diagram. Tell whether each pair of lines is parallel, perpendicular, or intersecting only.

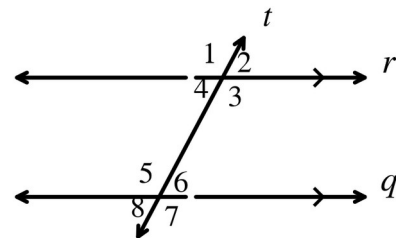
- a) line a and line b
- b) line a and line c
- c) line b and line d
- d) line c and line d



Example 3 - Angle Pairs Formed by a Transversal

Lines q and r are cut by transversal t. Give one example of each angle pair.

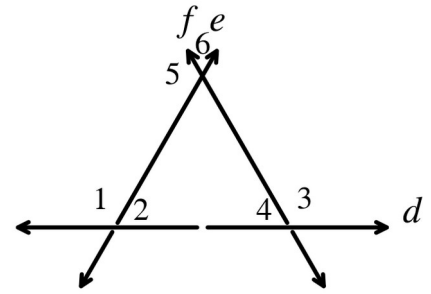
- a) corresponding angles
- b) alternate interior angles
- c) alternate exterior angles
- d) consecutive (same-side) interior angles



Example 4 - Identifying a Transversal

Lines d , e , and f form a triangle. Answer each question.

- Which line is a transversal of lines e and f ?
- Which line is a transversal of lines d and f ?
- Using d as the transversal, classify $\angle 1$ and $\angle 4$.
- Using d as the transversal, classify $\angle 2$ and $\angle 4$.



Example 5 - Always, Sometimes, or Never?

Decide whether each statement is always, sometimes, or never true. Explain.

- Two lines that do not intersect are parallel.
- Two lines that are perpendicular to the same plane are parallel.
- A transversal intersects two lines at the same point.